

Ishpreet Nagi

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EDUCATION

McMaster University

Bachelor of Applied Science, Computer Science

Hamilton, ON

Sept 2021 — May 2026

- Cumulative GPA: Summa Cum Laude | Dean's Honor List
- Relevant Coursework: Data Structures, Algorithms, Databases, Operating Systems, Concurrency, Computer Architecture, Cryptography, Discrete Mathematics

WORK EXPERIENCE

Software Engineer and Research Coordinator

June 2024 — July 2025

Healthcare Systems Research & Analysis

Waterloo, ON

- Designed distributed backend data pipelines in **Python** with **PostgreSQL**, Pandas, and NumPy to ingest, preprocess, and validate large interconnected datasets, **automating 90% of downstream deployment**
- Implemented RESTful service interfaces and background job workers for long-running processing, with retry logic, schema validation, and clear technical documentation for downstream consumers
- Tuned relational queries and schema design (indexing, joins, batch reads) to keep multi-million-row workloads responsive, while leading 4 engineers across 8 stakeholders to on-time delivery

Machine Learning Analyst and Research Assistant

May 2023 — May 2025

McMaster University - Department of Computing and Software

Hamilton, ON

- Built and maintained backend evaluation and model-serving infrastructure in **Python** under Dr. Douglas Down, achieving **10% accuracy gains** with 15% smaller models
- Independently maintained a research codebase across 2+ years — versioned schemas, reproducible pipelines, CI checks, and clear docs — enabling others to extend it without hand-holding

PROJECTS

Kollec App (kollec.app)

Sept 2025 — Apr 2026

- Designed and shipped scalable REST APIs and microservices using **Next.js API routes**, **Prisma ORM**, and **PostgreSQL**, owning the backend lifecycle from schema design through production deployment
- Built a stateful peer-to-peer trade-matching engine backed by complex relational queries and asynchronous updates across users, collections, and offers, applying DS&A choices to keep latency low at scale
- Implemented query optimization and indexing strategies (composite indexes, pagination, N+1 elimination) with structured logging so slow paths stay visible in production
- Built an event-driven workflow for asynchronous card-identification jobs, hardened with retries, idempotency keys, and technical docs for internal API consumers
- Maintained CI/CD pipelines with **Git**, **GitHub Actions**, and **Docker** within a team of 7, driving code review and test-coverage standards

Technical Executive, DeltaHacks 10 (github.com/deltahacks/landing-10)

Aug 2023 — May 2024

- Independently maintained the DeltaHacks 10 platform over a full cycle, shipping to **1000+ applicants** under a hard deadline
- Built instrumentation and structured activity logging for user engagement, treating telemetry as a first-class backend concern

SKILLS

- **Languages:** Python, Java, TypeScript, JavaScript, SQL (PostgreSQL, MySQL), C/C++ (familiar)
- **Backend & APIs:** REST, RPC (familiar), microservices, Node.js, Spring Boot (familiar), Prisma ORM, event-driven architecture, system design
- **Data Stores:** PostgreSQL, MySQL, Supabase, Redis (familiar), Elasticsearch (familiar), query optimization, schema design, indexing
- **Infra & AI Tooling:** Distributed systems, concurrency, Kafka/Flink (interest), AWS (interest), Docker, GitHub Actions (CI/CD), Git, GitHub Copilot, Cursor, Claude Code, LLM integration